INCH-POUND

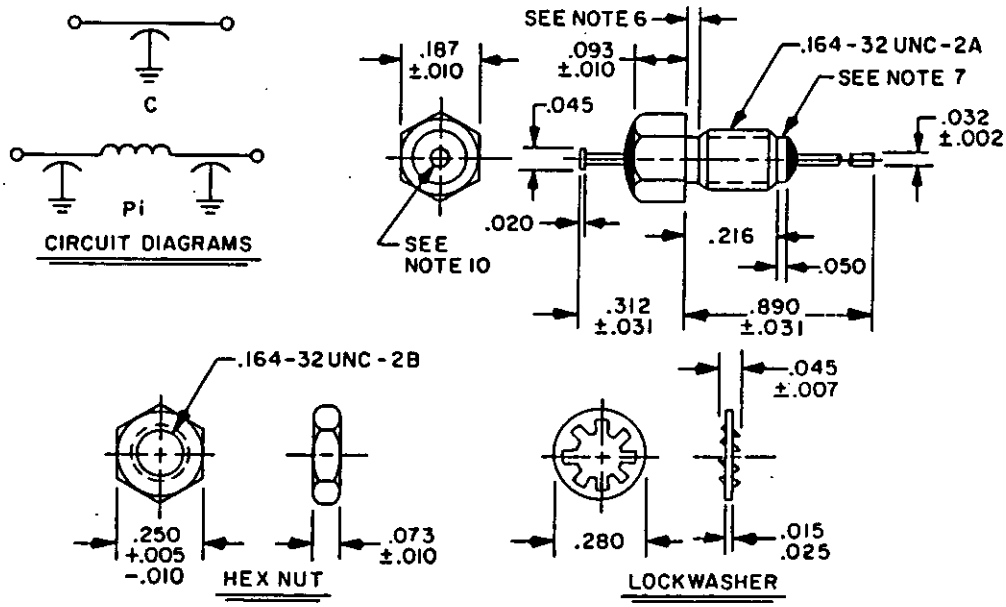
MIL-F-28861/7C
2 June 1989
SUPERSEDING
MIL-F-28861/7B
31 December 1987

MILITARY SPECIFICATION SHEET

FILTERS AND CAPACITORS, RADIO FREQUENCY/ELECTROMAGNETIC INTERFERENCE SUPPRESSION, NONHERMETICALLY SEALED, STYLE FS61

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-28861.



Inches	mm	Inches	mm	Inches	mm
.002	0.05	.030	0.76	.164	4.17
.005	0.13	.031	0.79	.187	4.75
.007	0.18	.032	0.81	.216	5.49
.010	0.25	.040	1.02	.250	6.35
.013	0.33	.045	1.14	.280	7.11
.015	0.38	.050	1.27	.312	7.92
.020	0.51	.073	1.85	.890	22.61
.025	0.64	.093	2.36		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005.
4. Circuit diagram is for information only.
5. All filters shall be supplied with mounting hardware (hex nut and lockwasher) Mounting hardware shall be furnished with the same finish as the filter case.
6. Imperfect thread or undercut optional .040 inch maximum.
7. One imperfect thread allowed .040 inch maximum.
8. Recommended mounting torque: 64 oz-in. ±4 oz-in.
9. Potting shall not extend beyond .030 inch from the filter body.
10. Turret head terminal optional.

FIGURE 1. Case and hardware dimensions and circuit diagrams.

© denotes changes

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 1.5 gram maximum.

Case finish: S (silver plated).
In accordance with MIL-F-28861.

Terminals: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table I.

Rated current: 10 amperes, maximum.

Capacitance to ground: See table I for capacitance value.

Dissipation factor: 3 percent maximum.

Voltage and temperature limits of capacitance: +15, -40 percent.

Insulation resistance:

At +25°C: 1000 megohm-microfarad or 100,000 megohms minimum, whichever is less.

At +125°C: 100 megohm-microfarad or 10,000 megohms minimum, whichever is less.

Insertion loss:

At +25°C: In accordance with table I.
At -55°C and +125°C: A 3 dB degradation from the +25°C value shall be allowed.

Voltage drop: 0.1 volt, maximum.

DC resistance: 0.01 ohm, maximum.

Seal: Not applicable.

Temperature rise: +25°C, maximum.

Solderability of terminals: In accordance with MIL-F-28861.

Product assurance level: Class B only.

© Part or Identifying Number (PIN): The Part or Identifying Number shall be as follows:

M28861/07-XXX-S-B

XXX: Dash no. per figure 1 and table I

S: Case finish

B: Product assurance level per MIL-F-28861

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TABLE I. Electrical characteristics.

Dash number	Circuit	Rated voltage dc volts	Capacitance (µF) -0, +100 percent	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/ 2/					
				1 MHz	10 MHz	100 MHz	200 MHz	1 GHz	10 GHz
001	C	50	.05	15	35	42	50	70	70
002	Pi	50	.032	12	30	60	70	70	70
003	C	100	.01	4	21	35	40	55	60
004	Pi	100	.012	-	12	60	70	70	70
005	C	100	.027	10	30	39	45	65	70
006	C	200	1,000 pF	-	4	20	25	40	50
007	Pi	200	2,000 pF	-	8	42	58	70	70
Ⓒ 008	C	200	5,000 pF	-	15	34	41	50	55

1/ For C circuits, insertion loss measurements shall be made under no load. For Pi circuits, insertion loss measurements shall be made under full load over the frequency range of 1 MHz to 10 MHz; above this frequency range, insertion loss shall be measured under no load.

2/ The insertion loss requirements between any two adjacent specified frequencies shall be that of the lower of the two frequencies in order to accommodate resonant dips.

Ⓒ Marking: Filters shall be marked, as a minimum, with an abbreviated Part or Identifying Number as shown on figure 2. Full marking, in accordance with MIL-F-28861, shall be marked on the unit package.

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J - Abbreviation for JAN
28861 - Abbreviation for M28861
B - Indicates specification sheet 07
C - Code indicating dash no. as follows:

Code	Dash no.	Code	Dash no.
A	001	E	005
B	002	F	006
C	003	G	007
D	004	H	008

B - Code indicating product assurance level (for example, B indicates class B).
221 - Date code. First digit indicates the year (for example, 2 indicates 1982). The last two digits indicate the week.

FIGURE 2. Example of marking for the abbreviated Part or Identifying Number on the hex flats - expanded view.

Cataloging information: Dash numbers 001, 003, 005, 006, and 008 shall be cataloged under FSC 5910 as feed-through ceramic capacitors. Dash numbers 002, 004, and 007 shall be cataloged under FSC 5915 as radio frequency interference filters.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, CG, MC, OS, SH
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 59GP-0081-02)

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